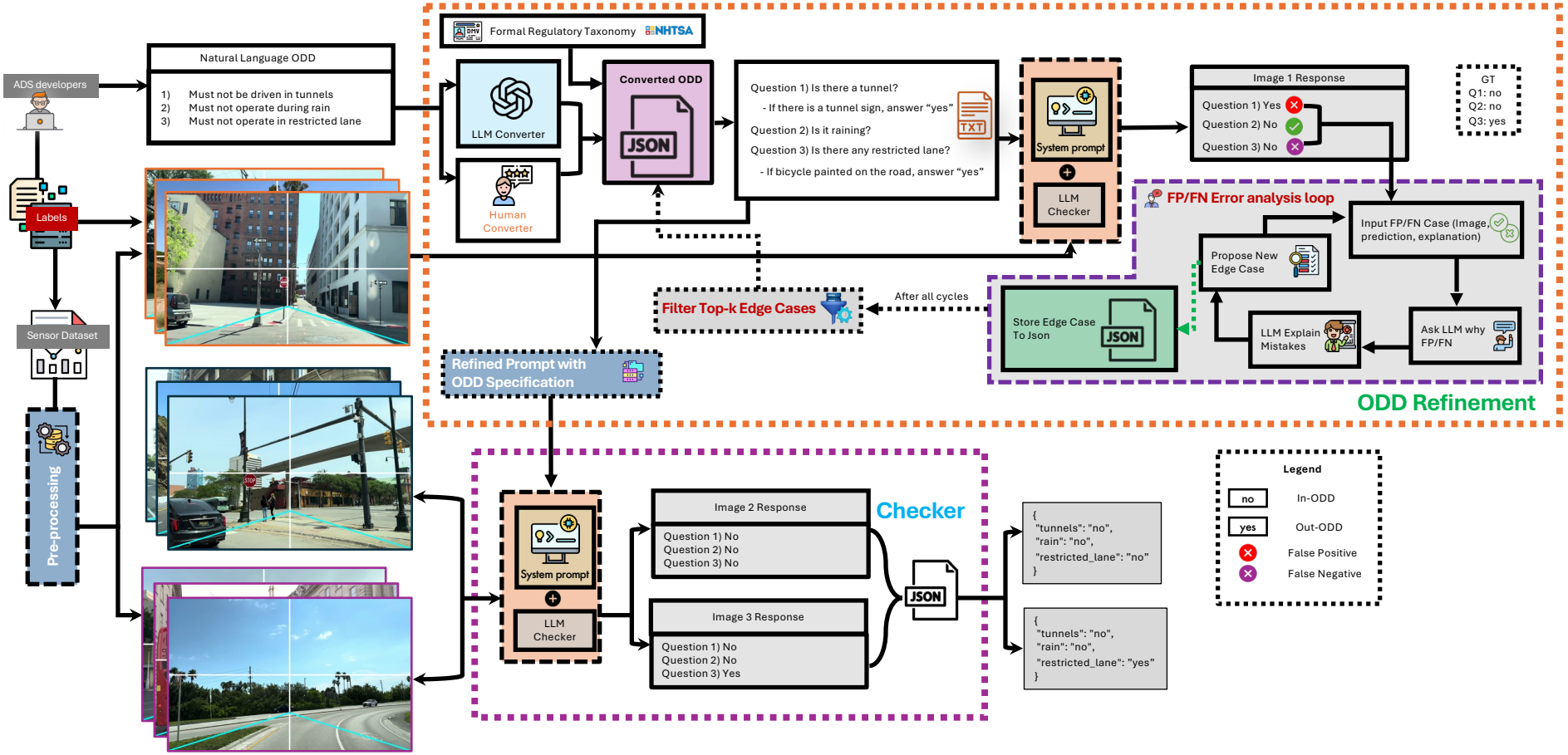

ADS developers
Natural Language ODD
1) Must not be driven in tunnels
2) Must not operate during rain
3) Must not operate in restricted lane
Formal Regulatory Taxonomy
NHTSA
LLM Converter
Human Converter
Converted ODD
JSON
Question 1) Is there a tunnel?
- If there is a tunnel sign, answer "yes"
Question 2) Is it raining?
Question 3) Is there any restricted lane?
- If bicycle painted on the road, answer "yes"
TXT
System prompt
LLM Checker
Image 1 Response
Question 1) Yes
Question 2) No
Question 3) No
GT
Q1: no
Q2: no
Q3: yes
FP/FN Error analysis loop
Input FP/FN Case (Image, prediction, explanation)
Propose New Edge Case
Store Edge Case To Json
JSON
LLM Explain Mistakes
Ask LLM why FP/FN
After all cycles
Filter Top-k Edge Cases
ODD Refinement
Labels
Sensor Dataset
Pre-processing
Refined Prompt with ODD Specification
System prompt
LLM Checker
Image 2 Response
Question 1) No
Question 2) No
Question 3) No
Image 3 Response
Question 1) No
Question 2) No
Question 3) Yes
Checker
JSON
{
"tunnels": "no",
"rain": "no",
"restricted_lane": "no"
}
{
"tunnels": "no",
"rain": "no",
"restricted_lane": "yes"
}
Legend
no
In-ODD
yes
Out-ODD
False Positive
False Negative